

Data Collection, Processing, and Engineering

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2. 1

Data Collection for AI

"Garbage in, garbage out — the data you collect determines the AI you build."

Data is the raw material of every AI system. Before a model can learn anything, you need to decide **what data to collect, how to collect it, and how to verify it is good enough**. This section walks through the key decisions every AI project must make at the very start.

WHY DATA COLLECTION MATTERS

THE BIG IDEA

Think of data collection like buying ingredients before cooking. If you walk into the kitchen with the wrong ingredients — or spoiled ones — no amount of cooking skill will save the meal. AI works the same way: the best algorithm on bad data still produces bad results.

FOUR KEY DATA CHARACTERISTICS

■ Completeness All required fields are present. Missing values create blind spots in the model.	■ Accuracy Values reflect reality. A blood pressure reading of 500/300 is impossible — flag it.
■ Consistency Same information stored the same way across all records and sources.	■ Timeliness Data is fresh enough for the task. Yesterday's stock price can't predict today's trade.

EXISTING DATASET VS. BUILDING YOUR OWN

Approach	Best When	Watch Out For
Use Existing Dataset (Kaggle, UCI, EHR databases)	You need to move fast; a public dataset covers your problem domain	Data may not match your context; potential licensing or privacy limits
Build Custom Dataset (surveys, sensors, scraping)	No existing dataset fits; your problem is highly specific	Expensive, time-consuming; risk of collection bias if design is poor

AUTOMATED VS. MANUAL COLLECTION

Method	Key Advantage	Key Limitation	Healthcare Example	Finance Example
Automated (APIs, sensors, scrapers)	Speed, scale, 24/7 operation	May collect irrelevant or low-quality data; requires technical setup	Vital-sign monitors logging patient readings every 5 minutes into EHR	APIs pulling real-time stock prices from exchanges
Manual (annotation, surveys)	Nuance, expert judgment, context	Slow and expensive; subject to human error	Radiologists marking tumor regions in mammograms	Analysts categorizing news articles for market sentiment

4-STEP STRATEGIC FRAMEWORK

1	Define Your AI Goal Write it in one clear sentence. Who is it for? What is it predicting? In what timeframe?
2	Analyze Resource Constraints Time, budget, expertise, legal compliance. Be realistic before committing to a collection approach.
3	Evaluate Data Requirements What type of data? What volume? How current must it be? Does it need to be labeled?
4	Choose Your Collection Strategy Existing vs. custom, automated vs. manual, or a hybrid — based on steps 1–3.

COMMON PITFALLS

■ "More Data = Better AI" Wrong. 10,000 high-quality relevant records often outperform 1,000,000 noisy, off-topic ones.	■ "Perfect Data" Paralysis Waiting for perfect data means never starting. Iterate and improve.
■ Data Hoarding Collecting everything "just in case" creates storage, cost, and privacy/compliance risks.	■ Automation Assumption Automated collection gives scale, not intelligence. Human judgment catches context that scrapers miss.

Section 2.1 — 5 Practice Questions

1. A healthcare team wants to train an AI to predict patient readmission. They find a public hospital dataset but it only covers patients aged 18–45. What data collection problem does this represent?

- A Automation assumption error
- B Lack of representativeness — the dataset does not cover all age groups relevant to the problem**
- C The "perfect data" paralysis trap
- D Timeliness issue — the data is too old

✓ **Correct Answer: B**

A dataset limited to one age group cannot produce a model that works well for older or younger patients. This is a representativeness problem — the data does not reflect the full population the AI will serve.

Teacher note: Distractor A targets confusion with automation errors. Distractor D tests whether students can distinguish representativeness from timeliness.

2. Which TWO of the following are advantages of automated data collection over manual collection? (Choose TWO)

- A It can operate continuously without fatigue**
- B It provides better judgment for ambiguous or contextual data
- C It scales easily to handle larger data volumes**
- D It eliminates the need for privacy or legal compliance review

✓ **Correct Answers: A and C**

Automated systems never sleep (A) and handle large volumes consistently (C). They do NOT replace human judgment for nuanced data (B), and legal/privacy compliance is still required regardless of how data is collected (D).

Teacher note: B and D are common misconceptions. Emphasize that automation solves scale and consistency, not ethical or contextual challenges.

3. A bank decides to use an existing credit-scoring dataset from a third party instead of collecting its own data. What is the MOST important risk to evaluate first?

- A The dataset may be too large to store

B The dataset may not accurately represent the bank's own customer population

C The collection method may have been too slow

D The dataset was probably collected manually and is therefore unreliable

✓ **Correct Answer: B**

Third-party datasets reflect someone else's customers, geography, and time period. The bank must verify the dataset matches its own customer profile before trusting predictions it generates.

Teacher note: D targets a common misconception that manual collection is inherently less reliable. Reinforce that collection method ≠ reliability.

4. Which of the following best describes the "data hoarding" problem in AI projects?

A Collecting data that is accurate but not recent

B Relying entirely on automated collection tools

C Collecting far more data than needed, much of which is irrelevant or unused

D Using data from only one source instead of multiple sources

✓ **Correct Answer: C**

Data hoarding means collecting data "just in case" without a clear purpose. This wastes storage, increases costs, and can introduce privacy risks from holding unnecessary personal information.

Teacher note: Connect this to Section 2.4 (resource costs). Storing unused data has real financial and compliance costs.

5. A team is building an AI for diagnosing skin conditions. They plan to collect images. What should be the FIRST step in their data collection strategy?

A Begin scraping images from the internet immediately

B Define the AI goal clearly and identify what types of images and labels are actually needed

C Select the machine learning algorithm before collecting any data

D Start with the largest dataset available regardless of relevance

✓ **Correct Answer: B**

Strategy must precede action. Defining the goal first — what conditions to detect, what labels are needed, what patient populations to cover — prevents collecting the wrong data at scale.

Teacher note: *C is a common student error: algorithm selection comes AFTER understanding the data, not before. A is tempting but skips legal and quality considerations.*

2. 2

Data Quality Assessment

"Even the smartest detective will reach wrong conclusions with bad evidence."

Data quality assessment is the systematic process of examining data to understand its **condition, reliability, and suitability** for a specific AI task. Think of it as a health check-up for your data before you trust it to make real-world decisions.

THE 6 DIMENSIONS OF DATA QUALITY

■ 1. Completeness Are all required fields filled? Missing data = blind spots in your model.	■ 2. Accuracy Do values reflect real-world truth? A blood pressure of 500/300 is physiologically impossible.
■ 3. Consistency Is the same concept recorded the same way everywhere? "Doctor" vs. "Dr." vs. "Physician" is a consistency problem.	■ 4. Validity Does data follow business rules? A date of "February 30" is logically invalid.
■ 5. Timeliness Is data fresh enough for the AI's purpose? 5-year-old housing prices cannot predict today's market.	■ 6. Uniqueness Are there duplicate records? One patient appearing twice can distort model training.

REAL-WORLD IMPACT OF POOR DATA QUALITY

Domain	Data Quality Problem	Real-World Impact
Healthcare (Hospital Readmission)	25% missing discharge codes; impossible vital signs; inconsistent drug names	Medication errors due to poor data quality occur in ~15% of hospital admissions
Finance (Customer Churn)	40% missing income data; 3 different customer ID formats; 8% duplicate records	Model targets wrong customers, wasting retention budget on people who were not actually at risk

8-STEP ASSESSMENT PROCESS

1	Initial Data Exploration Profile the data: row count, column types, value ranges, and basic statistics.
2	Completeness Assessment Identify and quantify missing values. Which fields are critical vs. optional?
3	Accuracy Verification Cross-check values against known ranges, external sources, or domain rules.
4	Consistency Analysis Look for the same entity described differently across records or tables.
5	Validity Checking Apply business rules: Are all dates valid? Are all amounts positive? Are all codes in approved lists?
6	Timeliness Evaluation Assess how recent the data is relative to the AI's purpose.
7	Uniqueness Analysis Detect and resolve duplicate records that could skew model training.
8	Relevance Assessment Does this data actually contain the information needed to solve the stated problem?

BEST PRACTICES

■ Set Standards Before Collecting Define acceptable missing rates, valid value ranges, and required formats upfront — before collection begins.	■ Build Quality Gates Into Pipelines Automated checks at the point of data entry catch problems early, when they are cheap to fix.
■ Prioritize by Business Impact A 3% missing rate in a low-priority field is very different from a 3% missing rate in your target variable.	■ Document Everything Every quality issue, transformation, and rule should be documented for reproducibility and compliance.

EXAM PRACTICE

Section 2.2 — 5 Practice Questions

6. A dataset contains a patient record showing a birth date of "February 30, 1985." Which data quality dimension does this violate?

A Timeliness

B Completeness

C Validity

D Uniqueness

✓ Correct Answer: C

Validity means values follow real-world business or logic rules. February 30 is not a real date. The field is present (not a completeness issue) and not a duplicate (not uniqueness) — it is logically invalid.

Teacher note: Students often confuse validity with accuracy. Accuracy = does this reflect reality? Validity = does this follow a defined rule or constraint?

7. Which TWO actions are most important during a data quality assessment? (Choose TWO)

A Identifying missing values and quantifying how many records are affected

B Selecting the machine learning algorithm to use before examining the data

C Cross-verifying data values against known ranges or external reference sources

D Maximizing the size of the dataset by adding any available records

✓ Correct Answers: A and C

Quality assessment must quantify completeness (A) and verify accuracy through cross-validation (C). Algorithm selection (B) comes after quality assessment, and indiscriminately adding records (D) can introduce more quality problems.

Teacher note: D is the "more data always better" trap from Section 2.1. Use this as a cross-section connection point.

8. In a hospital database, the same medication "Acetaminophen" is stored as "acetaminophen," "Tylenol," and "APAP" across different departments. Which quality dimension is violated?

A Completeness

B Consistency

C Timeliness

D Accuracy

✓ Correct Answer: B

Consistency means the same concept is represented the same way throughout the dataset. All three names refer to the same drug, but an AI cannot know that without consistent encoding.

Teacher note: *D is a common trap — students assume multiple names means inaccuracy. Emphasize: all three names are factually correct; the problem is inconsistent representation.*

9. A financial firm uses 5-year-old customer transaction data to train a fraud detection model. Which data quality issue is most likely to affect model performance?

- A Uniqueness — duplicate transactions in the dataset
- B Timeliness — fraud patterns have likely changed significantly in 5 years**
- C Completeness — too many missing transaction amounts
- D Validity — transaction dates fall outside accepted ranges

✓ Correct Answer: B

Fraud techniques evolve rapidly. A model trained on 5-year-old patterns may not recognize modern fraud methods, even if the historical data is complete and accurate.

Teacher note: *Use this to introduce the concept of model drift in Section 4/5. Timeliness in training data directly feeds into why models degrade over time.*

10. What is the PRIMARY purpose of setting up automated quality monitoring in a data pipeline?

- A To replace the need for human data analysts entirely
- B To speed up model training by pre-processing data faster
- C To catch quality problems early before they propagate into model predictions**
- D To reduce the size of the dataset by removing records automatically

✓ Correct Answer: C

Automated quality gates catch problems (missing values, outliers, format violations) at the point of entry, preventing bad data from reaching the AI model and corrupting its predictions.

Teacher note: *A is a common AI-anxiety misconception. Emphasize that automation augments human oversight; it does not replace it.*

2. 3

Ensuring Representative Data in AI

"If you only teach a child about cats, they'll call every dog a 'strange cat.'"

Representative data means the dataset accurately reflects the **full diversity of real-world situations** the AI will encounter. Non-representative data produces AI systems that work well for some groups and fail — sometimes harmfully — for others.

THE THREE PILLARS OF REPRESENTATIVE DATA

■ Coverage The dataset includes all relevant subgroups: age groups, genders, ethnicities, income brackets, geographic regions.	■■ Balance Subgroups are proportionally represented. An 80/20 imbalance means the model is optimized for the majority.
■ Temporal Relevance Data reflects conditions at the time and place the AI will actually be used — not just where it was collected.	

TYPES OF COLLECTION BIAS

Bias Type	What Happens	Real-World Example
Selection Bias	Only certain groups are included in data collection	Training a skin cancer AI only on light-skinned patients — it fails on darker skin tones
Historical Bias	Past data reflects past discrimination, which the AI then perpetuates	Using historical loan approvals (which denied minorities more often) to train a new credit model
Measurement Bias	Data is collected differently for different groups	Hospital data over-represents insured patients; uninsured patients' outcomes are missing
Feedback Loop Bias	The AI's decisions create new data that reinforces the original bias	Predictive policing sends more officers to certain areas → more arrests → model confirms those areas are high crime

WHY THIS IS AN ETHICS ISSUE, NOT JUST A TECHNICAL ONE

When AI systems fail disproportionately for underrepresented groups — denying loans, misdiagnosing conditions, flagging innocent people — the consequences fall on those groups. Representative data is not optional; it is an ethical obligation.

PROXY VARIABLES

Removing a sensitive column like "race" does not guarantee a fair model. Other variables can act as **proxy variables** — they appear neutral but effectively encode a protected characteristic. For example, zip code correlates strongly with race due to decades of discriminatory housing policies. An AI using zip code may discriminate by proxy even when race was never included in the dataset.

ADEQUATE DATA VOLUME: THE GOLDBLOCKS PRINCIPLE

Factor	Effect on Required Data Volume
Problem Complexity	More complex distinctions (early-stage cancer vs. obvious tumors) require more examples because differences are subtle.
Population Diversity	More subgroups = more data needed to ensure each subgroup is adequately represented.
The "Minority Report" Trap	Including 200 records from a minority group alongside 100,000 from a majority group does NOT solve representativeness — 200 is not enough to learn meaningful patterns.

4-STEP IMPLEMENTATION FRAMEWORK

1	Define Your Real-World Population Before collecting anything, enumerate every type of person, situation, and edge case your AI will encounter in production.
2	Identify Potential Bias Sources For every step in collection, ask: who might be excluded by this method? Does this tool assume technology access? Language proficiency? Geographic location?

3

Design Mitigation Strategies

Use multiple collection methods. Partner with organizations serving underrepresented communities. Actively seek out missing subgroups.

4

Implement Ongoing Quality Controls

Regular audits: is the distribution of your data still matching your real-world population? Monitor for new subgroups and emerging gaps.

**EXAM
PRACTICE****Section 2.3 — 5 Practice Questions**

11. A hospital trains a sepsis detection AI using only data from its ICU. When deployed in the general ward, the model performs poorly. What is the MOST likely cause?

- A The model used the wrong algorithm for the task
- B The training data was not representative of the patient population in the general ward**
- C The general ward patients had worse data quality than ICU patients
- D Sepsis detection requires manual collection methods only

✓ Correct Answer: B

ICU patients are sicker and monitored more intensively than general ward patients. A model trained on ICU data learns patterns specific to critically ill patients and cannot generalize. The deployment context was not represented in training data.

Teacher note: This is a classic example of domain shift — the difference between training distribution and deployment distribution.

12. What is a "proxy variable" in the context of AI bias?

- A A variable that substitutes for missing data in a dataset
- B A seemingly neutral variable that actually correlates with and encodes a protected characteristic**
- C A feature created during feature engineering to improve model accuracy
- D A backup variable used when the primary variable has too many missing values

✓ Correct Answer: B

Zip code, last name, and school attended are examples of proxy variables. They appear neutral but can encode race, ethnicity, or socioeconomic status, allowing a model to discriminate without explicitly using protected attributes.

Teacher note: *A is a plausible distractor since "proxy" sounds like a substitute. Emphasize the concept of correlation with protected characteristics.*

13. Which TWO of the following best describe strategies for improving data representativeness? (Choose TWO)

A Using stratified sampling to ensure all subgroups are proportionally included

B Removing all data from minority subgroups to reduce complexity

C Auditing the data collection process to identify where certain groups may be under-collected

D Training the model only on data where the target outcome is known with certainty

✓ **Correct Answers: A and C**

Stratified sampling (A) ensures proportional coverage. Auditing collection processes (C) surfaces systematic gaps before they become model biases. B would worsen representativeness; D describes labeled data selection, not representativeness.

Teacher note: *B tests whether students understand that removing minority data is the opposite of the solution.*

14. A predictive policing AI is deployed and begins sending more officers to Neighborhood X. More arrests occur there. The model then identifies Neighborhood X as higher risk. This is an example of:

A Measurement bias

B Sampling bias

C Feedback loop bias

D Historical bias

✓ **Correct Answer: C**

A feedback loop bias occurs when the AI's own decisions generate new training data that reinforces the original prediction. More policing → more arrests → model "confirms" the area is high-crime, regardless of actual crime rates.

Teacher note: *This is one of the most powerful examples of how an AI can perpetuate injustice while appearing objective. Use it as a discussion anchor.*

15. An AI company is building a loan approval model. They remove the "race" column from their dataset before training. Does this guarantee a fair model?

A Yes — removing protected attributes from training data always ensures fairness

B No — proxy variables like zip code or last name may still encode race indirectly

C Yes — as long as the model accuracy is high, the model is fair

D No — but only because historical data should never be used for lending decisions

✓ **Correct Answer: B**

Simply removing the "race" column does not prevent discrimination. Other variables (zip code, name, school attended) can act as proxies and effectively re-encode race, allowing the model to discriminate by proxy.

Teacher note: A is the most common student error. This is a critical takeaway: fairness requires more than removing sensitive columns.

2.4

AI Resource Requirements & Project Feasibility

"You wouldn't drive cross-country without checking the fuel, tires, and budget first."

Many AI projects fail not because the idea was bad — but because teams never asked: **"Do we actually have what it takes to build this?"** Resource assessment answers that question before money is spent.

THE FOUR RESOURCE DIMENSIONS

■ Computing Resources CPU/GPU processing power, cloud vs. on-premise. Some models require supercomputers; others run on a laptop.	■ Time Complexity How long will training take as data grows? A model that takes 2 days to train cannot support real-time decisions.
■ Data Requirements Volume, variety, and availability. Some problems need millions of labeled examples; others can work with hundreds.	■ Budget & Expertise Cloud compute, data labeling, talent, and maintenance all carry real costs that must be planned upfront.

WHEN TO PROCEED VS. PAUSE

Signal	Likely Decision
Problem is well-defined, data exists, and compute is accessible	■ Proceed
Problem is clear but training data does not exist and would be expensive to collect	■■ Pause — assess collection cost vs. business value
Required model would take months to train on available hardware	■■ Consider cloud compute or a simpler model
Similar AI solutions already exist and can be purchased	■ Buy vs. Build analysis required
Regulatory requirements for the domain are unclear	■ Delay — compliance must be confirmed first

REAL-WORLD EXAMPLE: DRUG DISCOVERY

Pharmaceutical companies have invested billions in AI-driven drug discovery. The compute requirements (training on molecular structures) are enormous, timelines run years, and the regulatory path adds years more. An organization without access to specialized hardware, rare chemistry datasets, and domain experts cannot replicate this — regardless of algorithmic sophistication.

EXAM PRACTICE

Section 2.4 — 5 Practice Questions

16. A startup wants to build an AI system for real-time fraud detection. The system must process each transaction in under 100 milliseconds. Which resource dimension is MOST critical to assess first?

- A Data budget — how much labeled fraud data they can afford
- B Time complexity and computing resources — the model must respond fast enough in production**
- C Team size — whether they have enough data scientists
- D Algorithm type — whether to use supervised or unsupervised learning

✓ Correct Answer: B

Real-time requirements impose strict latency constraints. If the model cannot produce a result in under 100ms, it is unusable — regardless of accuracy. This makes compute and time complexity the primary feasibility gate.

Teacher note: D is a common distractor — students often jump to algorithm selection. Emphasize: infrastructure feasibility comes before algorithm design.

17. Which TWO of the following are considered "hidden costs" that teams often underestimate in AI projects? (Choose TWO)

- A Ongoing model monitoring and maintenance after deployment**
- B The initial cost of selecting a cloud provider
- C Data labeling — the human time required to annotate training examples**
- D The cost of buying a new laptop for the team

✓ Correct Answers: A and C

Post-deployment monitoring (A) is a recurring cost teams frequently forget to budget. Data labeling (C) — paying humans to annotate images, text, or audio — can cost more than compute for supervised learning projects.

Teacher note: B and D are visible, obvious costs. The exam tests whether students identify the non-obvious costs.

18. A hospital AI team realizes their on-premise servers would take 6 months to train the model they need. What is the MOST practical next step?

- A Cancel the project — if on-premise hardware is insufficient, the project is not feasible
- B Reduce the training dataset until the model trains in a week
- C Evaluate cloud computing options (AWS, Google Cloud, Azure) for scalable GPU access**
- D Switch to unsupervised learning, which always trains faster

✓ **Correct Answer: C**

Cloud computing allows teams to rent powerful GPU clusters for short periods, solving a hardware bottleneck without purchasing new infrastructure. A is defeatist. B may compromise model quality. D is incorrect — learning type does not determine training speed.

Teacher note: D tests a common misconception about learning type vs. compute requirements.

19. Which of the following best describes why "time complexity" matters for AI project feasibility?

- A It determines how long it takes to hire the right data science team
- B It describes how processing time increases as data volume grows, affecting both training and real-time inference**
- C It measures how much time the end user spends interacting with the AI
- D It only applies to models with more than 1 million training examples

✓ **Correct Answer: B**

Time complexity is a measure of how computational demands scale with data size. A model with poor time complexity may be fine on small data but prohibitively slow at scale or for real-time use.

Teacher note: D is a common student assumption. Time complexity is relevant at any scale.

20. A team wants to build an AI for detecting rare diseases. They find that only 50 confirmed cases exist in the world. What is the PRIMARY feasibility concern?

- A Computing resources — rare disease detection requires specialized GPUs
- B Budget — rare disease datasets are always very expensive to purchase

C Data volume — 50 cases is almost certainly insufficient to train a reliable supervised learning model

D Algorithm selection — classification algorithms cannot handle rare diseases

✓ **Correct Answer: C**

Supervised learning models typically need hundreds to thousands of labeled examples to generalize. 50 cases creates a class-imbalance problem so severe that the model may simply learn to always predict "no disease" and achieve 99.9% accuracy while being clinically useless.

Teacher note: *This is a great bridge to the concept of class imbalance in Section 3 (algorithms and models).*

2.5

Data Conversion for AI

"Every photo, word, and sound must be translated into numbers before AI can learn from it."

AI systems are fundamentally mathematical. They can only process **numbers**. Everything else — text, images, audio, video, categorical labels — must be converted into numerical form before a model can learn from it.

THE THREE PILLARS OF DATA CONVERSION

■ 1. Format Standardization Convert all data into consistent, structured formats. Images must be the same size; dates must use the same format.	01 2. Binary Representation All information lives in computers as 0s and 1s. Images become pixel arrays; text becomes character codes.
■ 3. Feature Engineering Transform raw numbers into meaningful variables. Raw date → day of week + is_holiday + season.	

THE KITCHEN PREP ANALOGY

Feature engineering is like preparing ingredients before cooking. You do not throw whole vegetables into a blender — you wash, peel, chop, and portion them first. Similarly, raw data contains valuable information that must be extracted, reshaped, and portioned before an AI algorithm can work with it effectively.

4-STEP FEATURE ENGINEERING PROCESS

1	Data Understanding Examine raw data types, ranges, distributions. What information is present? What is missing?
2	Identify Transformations Determine what changes are needed: normalization, encoding, imputation, or new feature creation.
3	Implement Transformations Apply changes systematically and document every step. Consistency is critical.

Create & Manage Processed Datasets

Store cleaned, transformed features with full documentation for reproducibility.

KEY TRANSFORMATION TECHNIQUES

Technique	What It Does	Example
Normalization / Scaling	Converts values to a common scale so large numbers do not dominate	Age (0–100) and income (0–1,000,000) both scaled to 0–1
One-Hot Encoding	Converts categorical text into binary columns (one column per category)	"Blood type" A/B/AB/O → four binary columns
Missing Value Imputation	Fills missing values with a calculated substitute (mean, median, predicted)	Missing patient weight filled with average for that age group
Temporal Feature Extraction	Breaks a date/time into separate useful features	"2023-03-15" → day=Wednesday, is_weekend=False, quarter=Q1
Text Vectorization	Converts words into numerical vectors that capture meaning	Patient feedback → sentiment score (-1 to +1) + complaint category

CRITICAL RULE: TRANSFORMATION CONSISTENCY

Whatever transformation you apply to training data **MUST** be applied identically to all new incoming data. If you normalized ages using a 0–100 scale during training, you cannot use a 0–120 scale at prediction time. Inconsistency here causes silent, hard-to-detect model failures.

**EXAM
PRACTICE****Section 2.5 — 5 Practice Questions**

21. A dataset contains a "blood_type" column with values A, B, AB, and O. Which transformation is most appropriate before using this data in a machine learning model?

A Normalization — scale values between 0 and 1

B One-hot encoding — create a separate binary column for each blood type

C Imputation — fill missing values with the most common blood type

D No transformation needed — AI models can process text directly

✓ Correct Answer: B

Blood type is a categorical variable with no natural numeric order. One-hot encoding creates binary columns (A=1 or 0, B=1 or 0, etc.) without implying that AB is "greater than" A — which label encoding would incorrectly suggest.

Teacher note: D tests whether students know AI cannot process raw text. A tests confusion between normalization (for continuous numeric data) and encoding (for categorical data).

22. A model was trained using income data normalized to a 0–1 scale based on a maximum income of \$200,000. A new customer submits an application with an income of \$350,000. What is the MOST important action to take?

A Reject the application because the value is out of range

B Cap the income at \$200,000 before normalizing, without documenting the change

C Apply the same normalization parameters used during training consistently, and flag the out-of-distribution value

D Re-train the model with the new income value before processing the application

✓ Correct Answer: C

Transformation consistency requires using the original normalization parameters at inference time. The out-of-distribution value should be flagged for review, but the transformation process must remain consistent.

Teacher note: D introduces a dangerous anti-pattern: re-training on every new value is operationally impossible and would cause model instability.

23. Which TWO of the following are examples of temporal feature extraction? (Choose TWO)

A Extracting 'day of week' and 'is_holiday' from a timestamp

B Replacing missing dates with the median date in the dataset

C Calculating the number of days since a patient's last hospital visit

D Converting a date column to binary format for computer storage

✓ Correct Answers: A and C

Temporal feature extraction means deriving meaningful time-based variables from raw date/time data. Day of week (A) and days since last visit (C) are examples that capture patterns a raw timestamp cannot. B is imputation; D is binary conversion — different concepts.

Teacher note: D tests whether students confuse binary representation (the underlying computer storage layer) with feature engineering (the analytical layer).

24. What is data leakage in the context of feature engineering?

- A When private training data is accidentally shared with the public
- B When information from the future or from the target variable is accidentally included in features used during training**
- C When feature values slowly drift over time after deployment
- D When too many features are created and the model overfits

✓ Correct Answer: B

Data leakage means the model "sees" the answer during training — for example, including a "post-diagnosis" column when predicting whether a patient will be diagnosed. The model performs well in testing but fails in the real world.

Teacher note: A describes a privacy breach. C describes data drift. Both are plausible distractors that test broader course coverage.

25. A machine learning engineer creates 200 new features through feature engineering. Model accuracy on training data is excellent but drops sharply on real-world data. What is the MOST likely cause?

- A The model did not use enough binary representation
- B The training data was collected using automated methods instead of manual methods
- C Too many features caused the model to overfit — it learned noise in the training data instead of real patterns**
- D The feature engineering transformations were applied inconsistently

✓ Correct Answer: C

Overfitting occurs when a model learns the training data too specifically — including random noise — rather than underlying patterns. Creating hundreds of features without selection often causes this. Fewer, well-chosen features usually generalize better.

Teacher note: D is a plausible distractor. Clarify: D would cause detectable, consistent failure. C causes the model to appear to succeed during development before failing in production.

Chapter 2 Complete